

Two full data analysis using <code>glm()</code> - Handout

1 Exercise 1

- a) We estimate the model parameters under the two specifications (bernoulli for individual responses and binomial for grouped data) using the `glm()` command and store the results in the objects `baby.bern` and `baby.bin` respectively. Next, we can extract and compare the following values of interest.

- The Estimates

```
##           beta.bern  beta.bin
## (Intercept) -1.6127038 -1.6127038
## sexGirl     -0.3125528 -0.3125528
## foodBreast  -0.6692946 -0.6692946
## foodSuppl   -0.1725424 -0.1725424
```

The estimated coefficients are the same under the two specifications.

- The Standard Errors

```
##           stderr.bern stderr.bin
## (Intercept)  0.1124100  0.1124100
## sexGirl      0.1410396  0.1410397
## foodBreast   0.1530057  0.1530057
## foodSuppl    0.2055847  0.2055847
```

The same goes for the standard errors.

- The Deviances

```
##    dev.bern  dev.bin
## 1 1452.449  0.7219218
```

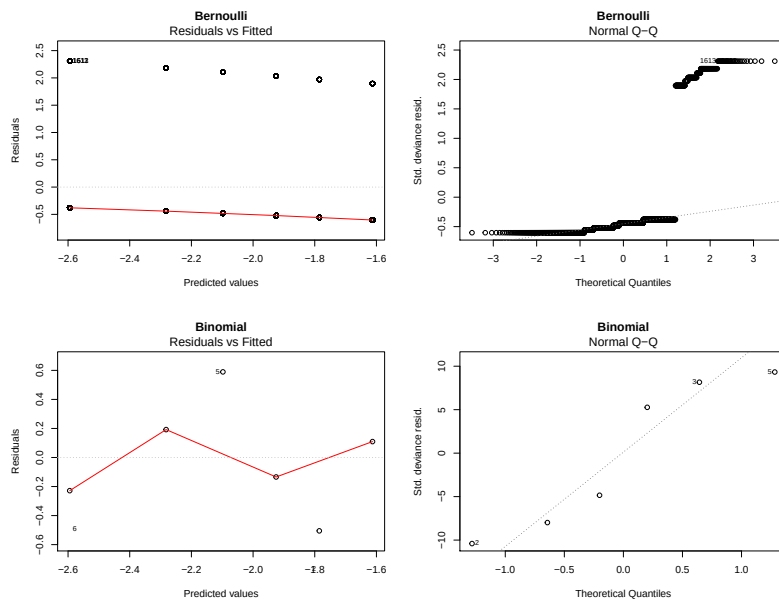
We see a difference.

- AIC Criteria

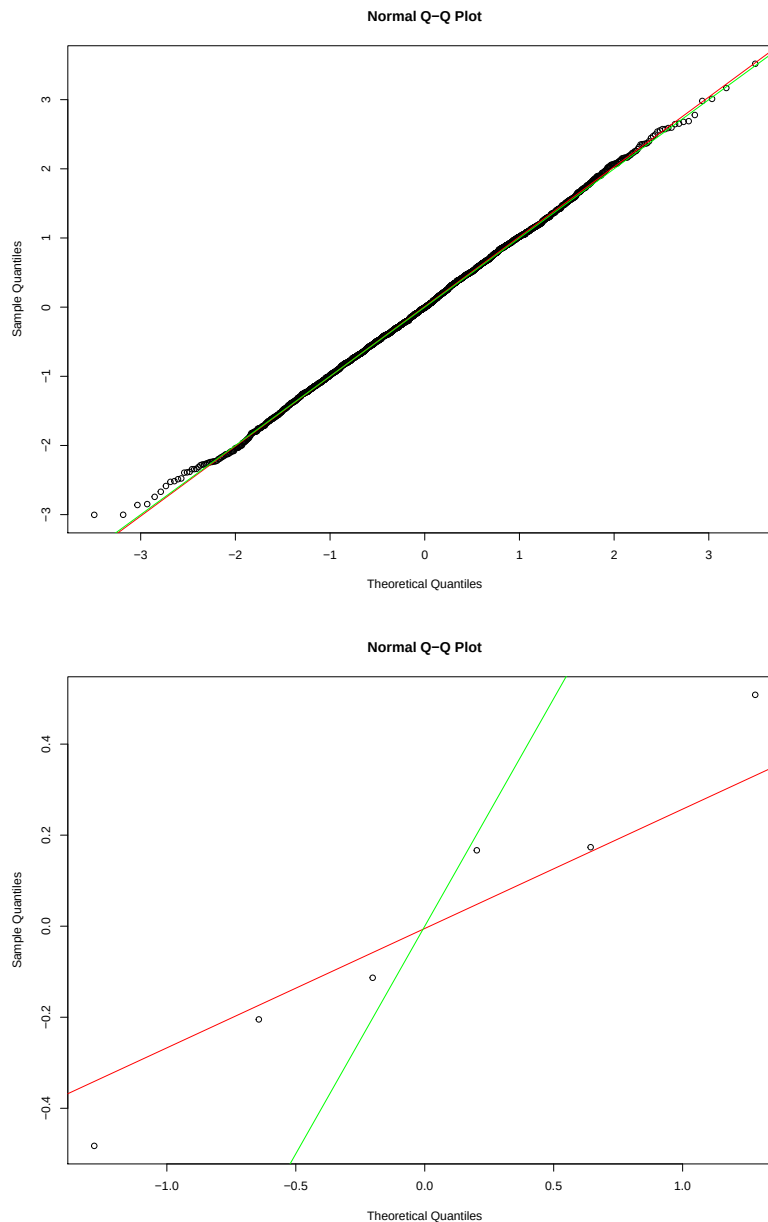
```
##      AIC.bern  AIC.bin
## 1 1460.449 40.23987
```

Again, we see a difference.

- Residual Plots



Residuals present the same type of structure as we would expect using both specifications, namely, bimodality of bernoulli residuals and approximate normality for binomial.



By using the randomized quantile residuals, it is apparent that the fit is satisfying for the bernoulli specification. Unfortunately, given the small amount of observations in the binomial case, it is hard to see whether this fit is any good under this other specification.

b) Remember that:

(a) $y_1, \dots, y_n \sim \mathcal{B}(1, p_i)$, with $n = 2074$,

$$\ell(\beta; \mathbf{y} | \mathbf{X}) = \sum_{i=1}^n y_i \log p_i + (1 - y_i) \log(1 - p_i) \quad (1)$$

(b) $Y_1, \dots, Y_6 \sim \mathcal{B}(m_i, p_i)$ with $\sum_{i=1}^6 m_i = n = 2074$.

$$\ell(\beta; \mathbf{Y} | \mathbf{X}) = \sum_{i=1}^6 \log \binom{m_i}{Y_i} + Y_i \log p_i + (m_i - Y_i) \log(1 - p_i) \quad (2)$$

Now, consider the log-likelihood for the Bernoulli model (1) and the grouping with respect to the six possible combinations of levels for variables **sex** and **food**:

$$y_{ij} \sim \mathcal{B}(1, p_i) \text{ for } j = 1, \dots, m_i \Rightarrow Y_i = \sum_{j=1}^{m_i} y_{ij} \sim \mathcal{B}(m_i, p_i) \quad (3)$$

We can thus rewrite, expression (1) as:

$$\ell(\beta; \mathbf{y} | \mathbf{X}) = \sum_{i=1}^n y_i \log p_i + (1 - y_i) \log(1 - p_i) \quad (4)$$

$$= \sum_{i=1}^6 \sum_{j=1}^{m_i} y_{ij} \log p_i + (1 - y_{ij}) \log(1 - p_i) \quad (5)$$

$$= \sum_{i=1}^6 \left(\sum_{j=1}^{m_i} y_{ij} \right) \log p_i + \left(\sum_{j=1}^{m_i} (1 - y_{ij}) \right) \log(1 - p_i) \quad (6)$$

$$= \sum_{i=1}^6 Y_i \log p_i + (m_i - Y_i) \log(1 - p_i) \quad (7)$$

$$\ell(\beta; \mathbf{y} | \mathbf{X}) = \ell(\beta; \mathbf{Y} | \mathbf{X}) - \sum_{i=1}^6 \log \binom{m_i}{Y_i} \quad (8)$$

i.e. the expression of the log-likelihood of a binomial model (2) minus the term accounting for the combinatory. Since optimization is performed by derivation with respect to β , the score equation and its second derivative will have the same expression, yielding the same results in terms of estimates and standard errors under both specifications.

- c) The expression of the Deviance in the Bernoulli model is as follows:

$$\mathcal{D}^*(\mathbf{y}, \hat{\mathbf{p}}) = 2 \sum_{i=1}^{2074} y_i \log \left(\frac{y_i}{\hat{p}_i} \right) + (1 - y_i) \log \left(\frac{1 - y_i}{1 - \hat{p}_i} \right) \quad (9)$$

However, due to the nature of the data ($y_i = 0$ or $y_i = 1$), we have that $y_i \log y_i = 0$ and $(1 - y_i) \log (1 - y_i) = 0$, yielding the following simplifications:

$$\begin{aligned} \mathcal{D}^*(\mathbf{y}, \hat{\mathbf{p}}) &= -2 \sum_{i=1}^{2074} y_i \log \hat{p}_i + (1 - y_i) \log (1 - \hat{p}_i) \\ &= -2 \sum_{i=1}^{2074} y_i \log \frac{\hat{p}_i}{1 - \hat{p}_i} + \log (1 - \hat{p}_i) \\ &= -2 \sum_{i=1}^{2074} y_i \mathbf{x}_i^T \hat{\boldsymbol{\beta}} + \log (1 - \hat{p}_i) \\ \mathcal{D}^*(\mathbf{y}, \hat{\mathbf{p}}) &= -2 \sum_{i=1}^{2074} \hat{p}_i \mathbf{x}_i^T \hat{\boldsymbol{\beta}} + \log (1 - \hat{p}_i) \quad \text{since : } \sum_{i=1}^{2074} (y_i - \hat{p}_i) \mathbf{x}_i = 0 \end{aligned} \quad (10)$$

From the last expression, it is apparent that, given the values of $\hat{\boldsymbol{\beta}}$, the deviance doesn't depend on y_i , making it a bad GOF measure. Consequently, it holds that the distribution of this statistic conditional on values of $\hat{\boldsymbol{\beta}}$ is degenerate, meaning that the approximation via the χ^2 is not possible.

In the binomial specification, this doesn't hold any more, given that Y_i can have different possible values, meaning that their contributions to the residual deviance are not neglected.

$$\mathcal{D}^*(\mathbf{Y}, \hat{\mathbf{p}}) = 2 \sum_{i=1}^6 Y_i \log \left(\frac{Y_i}{m_i \hat{p}_i} \right) + (m_i - Y_i) \log \left(\frac{m_i - Y_i}{m_i (1 - \hat{p}_i)} \right) \quad (11)$$

In that case, the statistic is a good GOF measure and its distribution can be approximated with the χ^2 . However, as in any case, the distribution of the deviance depends also on the number of available observations! Moral of the story : be careful when using the deviance in a binomial model.

- d) The AIC criterion is based on the Log-likelihood, meaning that it could be used for model selection under both specifications. However, when used, you should

be attentive on comparing two nested models.

2 Exercise 2

Check the code for an annotated solution.